Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Campbell, Melissa		
Project Title:	Development, Testing, and Verification of an Anaerobic Biological Water Processor		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	melissa.l.campbell1@jsc.nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-9718
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Mail Code EC3/HAM		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	01/01/2004
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	None provided. (Ed. note 1/14/2016: added from SAE Technical Paper abstract related to subject)--Biological water processors are currently being developed for application in microgravity environments. Work has been performed to develop a single-phase, gravity independent anoxic denitrification reactor for organic carbon removal. As a follow on to this work it was necessary to develop a gravity independent nitrification reactor in order to provide sufficient nitrite and nitrate to the organic carbon oxidation reactor for the complete removal of organic carbon. One approach for providing the significant amounts of dissolved oxygen required for nitrification is to require the biological reactor design to process two-phase gas and liquid in microgravity.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report received.
Bibliography Type:	Description: (Last Updated: 02/27/2004)